DNS Query Tools And SMB Enumeration

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Executive Summary

This lab focused on conducting
enumeration operation on target using DNS queries and SMB tools and thereby comparing the difference and weight in the returned responses. Key activities command of interest include nslookup, dig, host and enum4linux. The lab revealed that the enum4linux returned more elaborated information than the DNS query command.

1.0 Introduction

DNS query tools and SMB enumeration are reconnaissance techniques used to gather initial information about a network by identifying domain records, hosts, and services through DNS, and discovering shared resources, users, and system details exposed via the SMB protocol, helping analysts understand the environment before deeper security assessment.

2.0 Aim and objectives.

The Objectives of this lab is to;

- perform DNS queries using various tools to gather domain information
- use enum4linux to enumerate SMB shares and users on target system
- analyze the results to inform penetration testing efforts.

2.1 Tools

- nslookup
- Host
- Dig
- Enum4linux

3.0 Methodology

3.1 DNS Queries with nslookup, host and dig

- The linux terminal was open and a nslookup command was run against the target http://192.168.56.101
- Another command host was ran against the same target http:/:192.168.56.101
- The third command dig was ran against the target http://192.168.56.101 with the MX and TXT flag

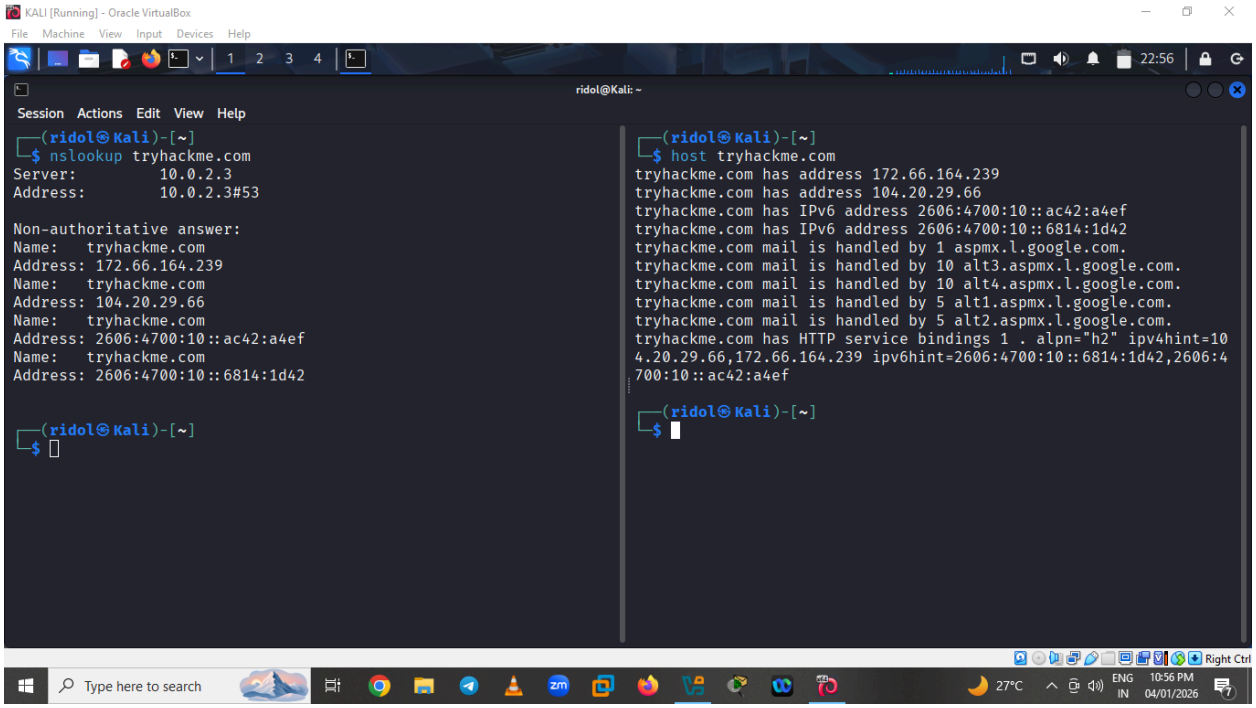

Fig 1: Evidence showing DNS queries of a target from linux terminal

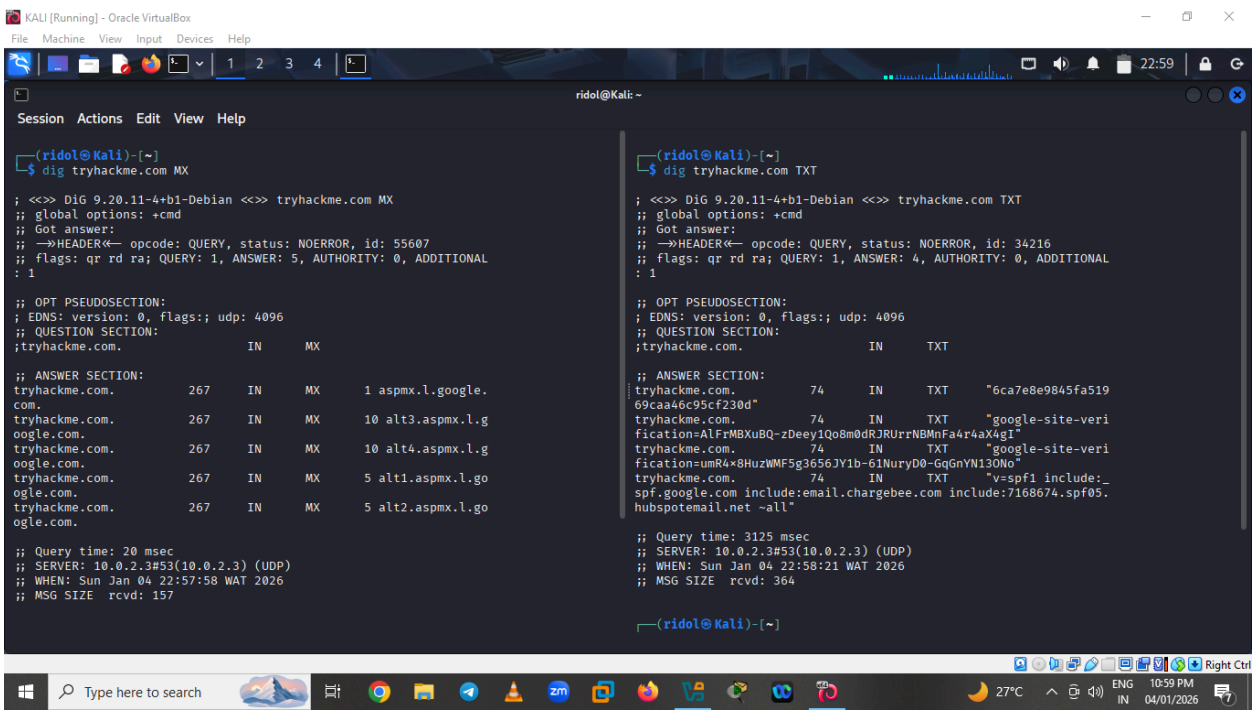

Fig 2: Evidence showing dig queries of a target from linux terminal

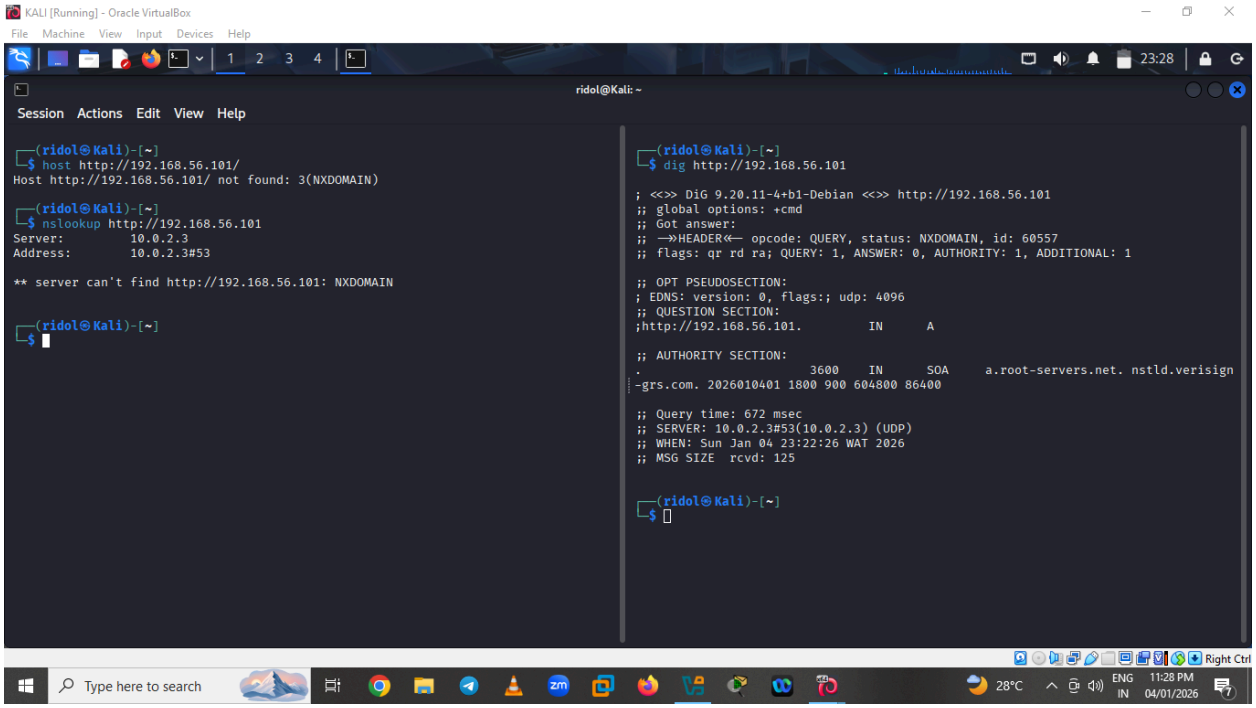

Fig 3: Evidence showing DNS queries of a target from linux terminal

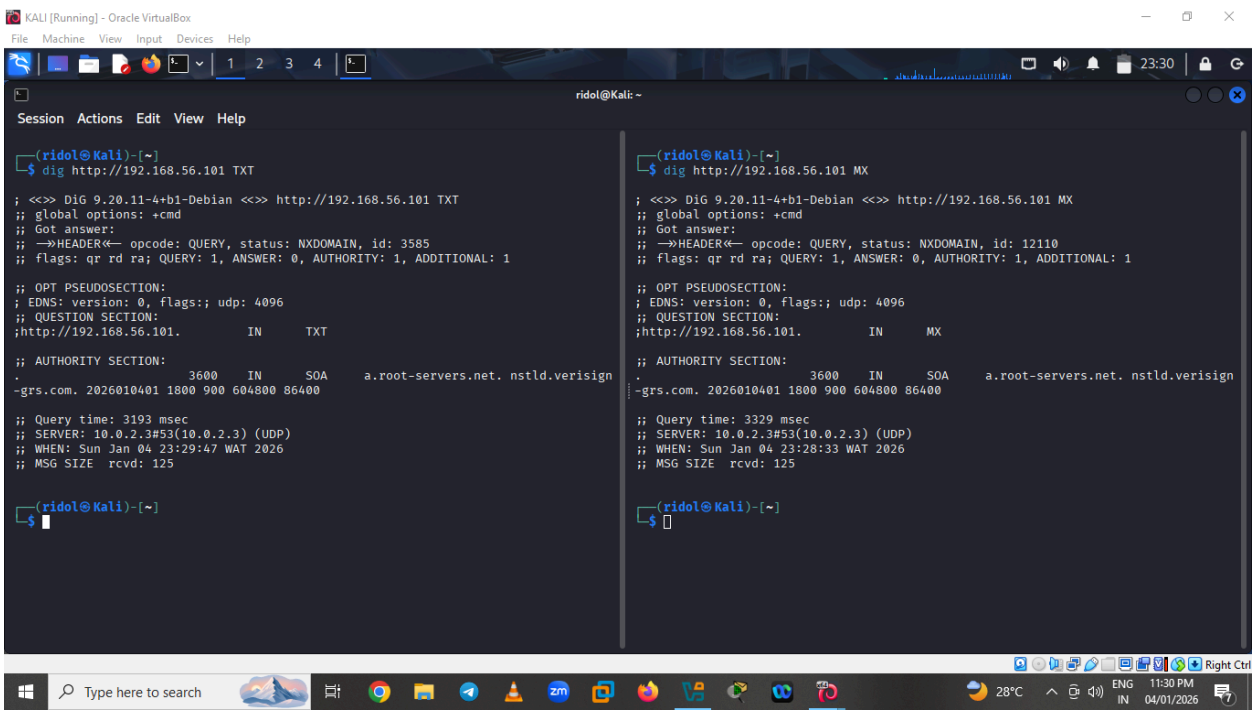

Fig 4: Evidence showing dig queries of a target from linux terminal

3.2 SMB Enumeration with Enum4linux

- The enum4linux application was installed using sudo apt update and sudo apt install enum4linux
- The command enum4linux -a 192.168.56.101 was ran on the linux terminal
- The same command was ran with a flag of -S and -u to filter out specific results from the response

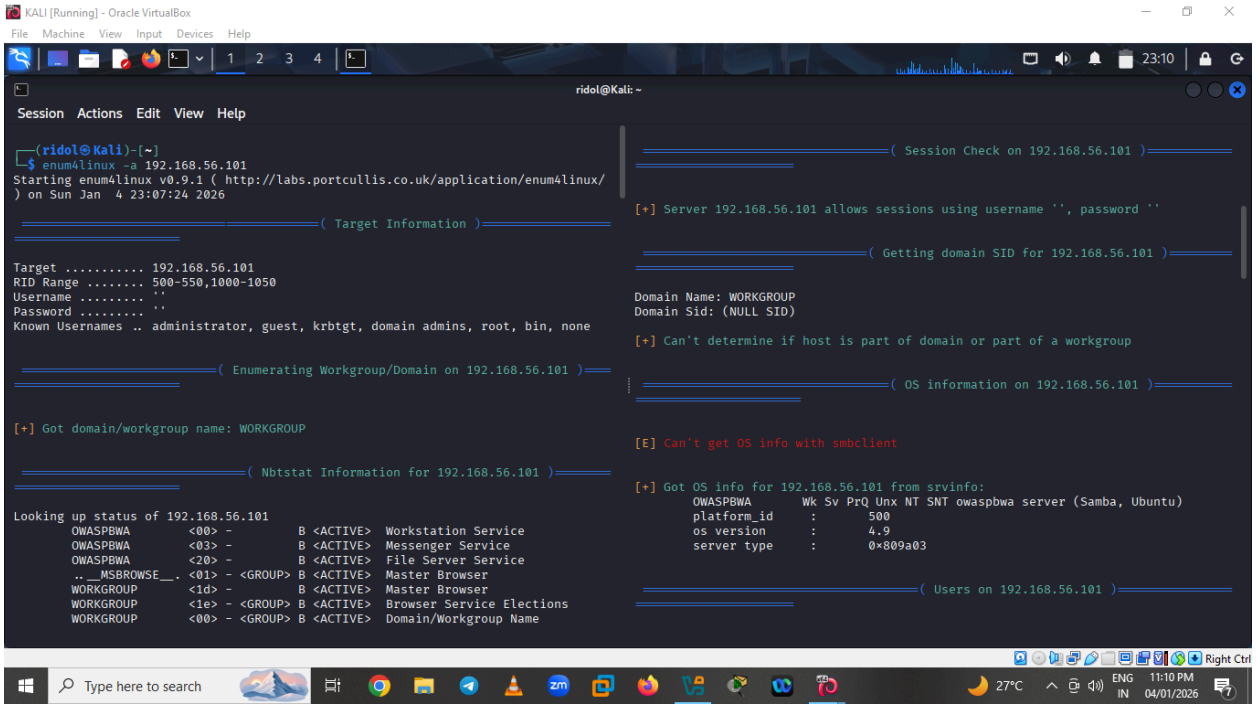

Fig 5: Evidence showing SMB response from enum4linux queries of a target from linux terminal

```
┌──(ridol㉿Kali)-[~]
└─$ enum4linux -S 192.168.56.101
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/
) on Sun Jan  4 23:17:35 2026

 ========================================( Target Information )========================================

Target ........... 192.168.56.101
RID Range ........ 500-550,1000-1050
Username ......... ''
Password ......... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none


 ============================( Enumerating Workgroup/Domain on 192.168.56.101 )============================


[+] Got domain/workgroup name: WORKGROUP


 ================================( Session Check on 192.168.56.101 )================================


[+] Server 192.168.56.101 allows sessions using username '', password ''


 ================================( Getting domain SID for 192.168.56.101 )================================

Domain Name: WORKGROUP
```

```
        ---------       ----      -------
        print$          Disk      Printer Drivers
        apache          Disk      Apache Web Server Root
        tomcat          Disk      Tomcat6 Root
        var             Disk      /var
        etc             Disk      /etc
        usr             Disk      /usr
        owaspbwa        Disk      /owaspbwa
        IPC$            IPC       IPC Service (owaspbwa server (Samba, Ubuntu))
Reconnecting with SMB1 for workgroup listing.

        Server               Comment
        ---------            -------

        Workgroup            Master
        ---------            -------
        WORKGROUP            OWASPBWA

[+] Attempting to map shares on 192.168.56.101

//192.168.56.101/print$ Mapping: DENIED Listing: N/A Writing: N/A
//192.168.56.101/apache Mapping: DENIED Listing: N/A Writing: N/A
//192.168.56.101/tomcat Mapping: DENIED Listing: N/A Writing: N/A
//192.168.56.101/var    Mapping: DENIED Listing: N/A Writing: N/A
//192.168.56.101/etc    Mapping: DENIED Listing: N/A Writing: N/A
//192.168.56.101/usr    Mapping: DENIED Listing: N/A Writing: N/A
//192.168.56.101/owaspbwa       Mapping: DENIED Listing: N/A Writing: N/A

[E] Can't understand response:

NT_STATUS_NETWORK_ACCESS_DENIED listing \*
//192.168.56.101/IPC$   Mapping: N/A Listing: N/A Writing: N/A
enum4linux complete on Sun Jan  4 23:18:11 2026
```

Fig 6: Evidence showing SMB response from enum4linux queries of a target from linux terminal

```
┌──(ridol㉿Kali)-[~]
└─$ enum4linux -u 192.168.56.101
enum4linux v0.9.1 (http://labs.portcullis.co.uk/application/enum4linux/)
Copyright (C) 2011 Mark Lowe (mrl@portcullis-security.com)

Simple wrapper around the tools in the samba package to provide similar
functionality to enum.exe (formerly from www.bindview.com).  Some additional
features such as RID cycling have also been added for convenience.

Usage: ./enum4linux.pl [options] ip

Options are (like "enum"):
    -U        get userlist
    -M        get machine list*
    -S        get sharelist
    -P        get password policy information
    -G        get group and member list
    -d        be detailed, applies to -U and -S
    -u user   specify username to use (default "")
    -p pass   specify password to use (default "")

The following options from enum.exe aren't implemented: -L, -N, -D, -f

Additional options:
    -a        Do all simple enumeration (-U -S -G -P -r -o -n -i).
              This option is enabled if you don't provide any other options.
    -h        Display this help message and exit
    -r        enumerate users via RID cycling
    -R range  RID ranges to enumerate (default: 500-550,1000-1050, implies -r)
    -K n      Keep searching RIDs until n consective RIDs don't correspond to
              a username.  Impies RID range ends at 999999. Useful
              against DCs.
    -l        Get some (limited) info via LDAP 389/TCP (for DCs only)
```

```
              This option is enabled if you don't provide any other options.
    -h        Display this help message and exit
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    -R range  RID ranges to enumerate (default: 500-550,1000-1050, implies -r)
    -K n      Keep searching RIDs until n consective RIDs don't correspond to
              a username.  Impies RID range ends at 999999. Useful
              against DCs.
    -l        Get some (limited) info via LDAP 389/TCP (for DCs only)
    -s file   brute force guessing for share names
    -k user   User(s) that exists on remote system (default: administrator,guest,k
rbtgt,domain admins,root,bin,none)
              Used to get sid with "lookupsid known_username"
              Use commas to try several users: "-k admin,user1,user2"
    -o        Get OS information
    -i        Get printer information
    -w wrkg   Specify workgroup manually (usually found automatically)
    -n        Do an nmblookup (similar to nbtstat)
    -v        Verbose.  Shows full commands being run (net, rpcclient, etc.)
    -A        Aggressive. Do write checks on shares etc

RID cycling should extract a list of users from Windows (or Samba) hosts
which have RestrictAnonymous set to 1 (Windows NT and 2000), or "Network
access: Allow anonymous SID/Name translation" enabled (XP, 2003).

NB: Samba servers often seem to have RIDs in the range 3000-3050.

Dependancy info: You will need to have the samba package installed as this
script is basically just a wrapper around rpcclient, net, nmblookup and
smbclient.  Polenum from http://labs.portcullis.co.uk/application/polenum/
is required to get Password Policy info.


┌──(ridol㉿Kali)-[~]
└─$
```

Fig 7: Evidence showing SMB response from enum4linux queries of a target from linux terminal

4.0 Results and Findings

The nsloolup command revealed basic information such the ipv4 and ipv6 mapped to the domain. The host command revealed more information than the nslookup command such as the domain mail handler.

The dig command showed additional information than both the previous command. This information includes the search header status, pseudosection such as version, flag, udp, aspmx, alt1-4.aspmx (for MX) and google site verification (for TXT).

Enum4linux returned more explicit information about the target. This information includes the domain workgroup and active services , information about the server and directories .

5.0 Conclusion

Results from SMB revealed detailed and elaborated information about the target far beyond the dns queries. Querying the ip address directly rather than the DNS gave the enum4linux a lot of edges as it does not have to engage in dns resolution before locating the target address which can be a fundamental problem to enumeration in case of dns resolution problem.